GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

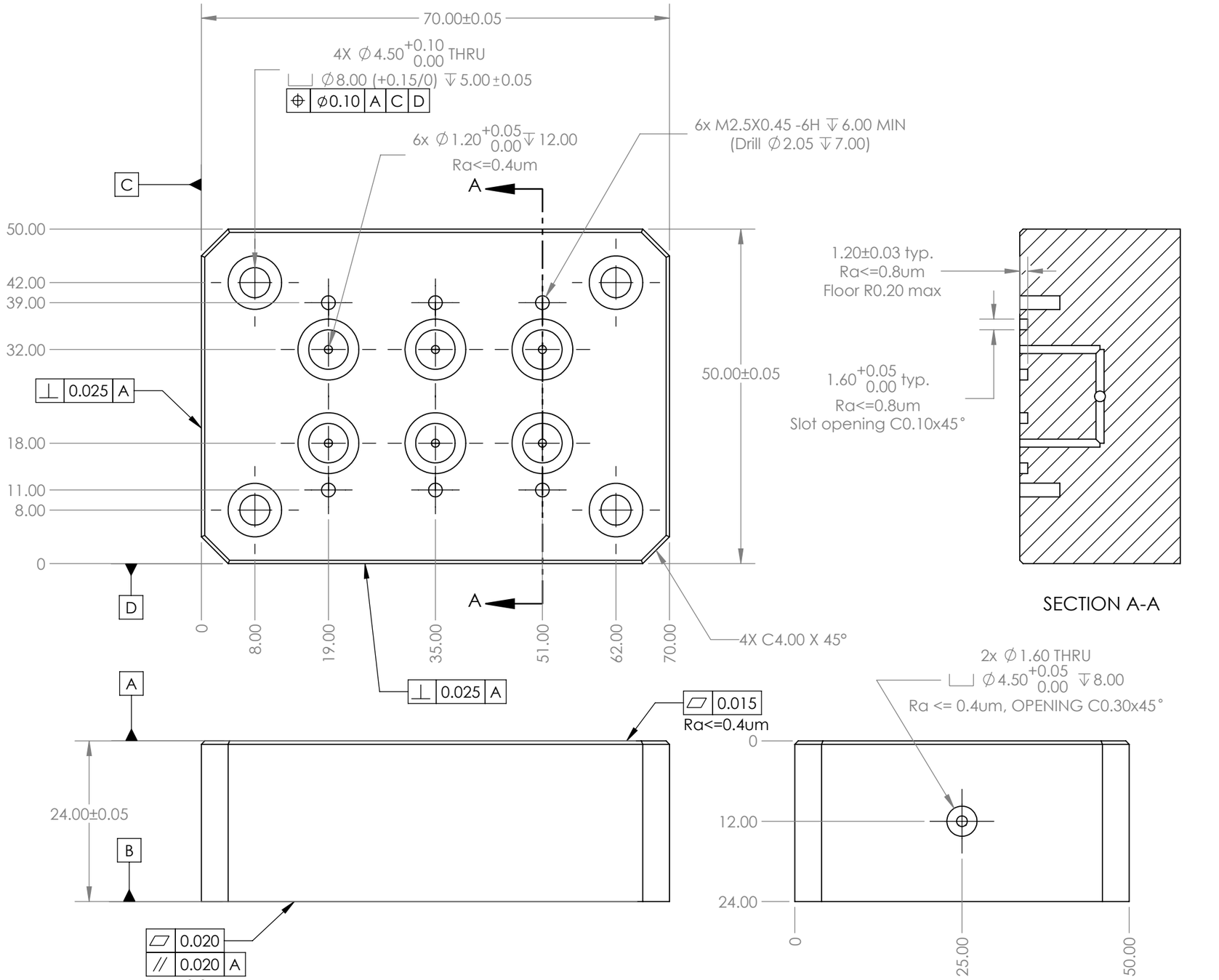
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-MED-IVD-MANIFOLD-5.3.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.2 DEG (HIGH-PRECISION MEDICAL CNC MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & BIO-COMPATIBILITY REQUIREMENTS:
2.1 MATERIAL OPTION A: MEDICAL GRADE VIRGIN PEEK (POLYETHERETHERKETONE) NATURAL COLOR, RESISTANT TO ACIDS, BASES, AND BIO-SURFACTANTS.
2.2 MATERIAL OPTION B: AUSTENITIC STAINLESS STEEL 316L (UNS S31603 / 1.4404), VACUUM MELTED, FULLY PASSIVATED PER ASTM A967 CITRIC ACID PROTOCOL.
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING OR RESIN 3D PRINTING PERMITTED TO AVOID PARTICULATE SHEDDING AND CONTAMINATION OF IN-VITRO DIAGNOSTIC REAGENTS.

3. MICROFLUIDIC PERFORMANCE & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (TOP VALVE MOUNTING FACE): FLATNESS <= 0.015 mm, SURFACE FINISH Ra <= 0.4 MICRONS. ZERO SCRATCHES, TOOL MARKS, OR VOIDS PERMITTED.
3.2 SECONDARY DATUM [B] (BOTTOM THERMAL MOUNTING FACE): FLATNESS <= 0.020 mm, PARALLEL TO [A] WITHIN 0.020 mm, Ra <= 0.8 MICRONS (SOLID BASE FOR TEC HEAT SINKING).
3.3 MICRO-CHANNELS (MAIN Ø1.60 mm, BRANCH Ø1.20 mm): CONTINUOUS, UNDERCUT-FREE, AND CAVITATION-FREE. INTERNAL CHANNEL WALL FINISH Ra <= 0.4 MICRONS (MICRO-HONED / POLISHED).
3.4 DEAD VOLUME CONSTRAINT: TOTAL ACCUMULATED CHANNEL DEAD VOLUME ACROSS ENTIRE MANIFOLD MUST BE LESS THAN 15 MICROLITERS (< 15 µL).

4. PRESSURE INTEGRITY & LEAKAGE TESTING:
4.1 PROOF PRESSURE TEST: 100% HELIUM / AIR-UNDER-WATER MASS SPECTROMETRY TEST AT 4.0 BAR (0.40 MPa) GAUGE PRESSURE FOR 60 SECONDS. ZERO BUBBLE / ZERO LEAKAGE PERMITTED (LEAK RATE < 1E-6 mbar·L/s).
4.2 CROSS-TALK / INTER-CHANNEL LEAKAGE: MINIMUM SOLID ISOLATION WALL THICKNESS BETWEEN ADJACENT MICRO-CHANNELS AND MOUNTING THREADS MUST BE >= 2.20 mm UNDER ALL TOLERANCE CONDITIONS.

5. CLEANLINESS PROTOCOL:
5.1 COMPONENT MUST BE CLEANED VIA MULTI-STAGE ULTRASONIC CAVITATION USING DEIONIZED WATER AND MEDICAL-GRADE SURFACTANT.
5.2 100% FREE OF CUTTING OILS, RESIDUAL SWARF, BURRS, AND PARTICULATES LARGER THAN 5 MICRONS (COMPLIANT WITH ISO 13485 / ISO 16232 MEDICAL APPARATUS CLEANLINESS).



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm Angular ±0.2° (Precision CNC Milling)	FINISH: Ultra-Clean Passivated (ASTM A967 for 316L) / Native Medical PEEK (No Chemical Leaching)				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION: A
	NAME	SIGNATURE	DATE			TITLE: DESKTOP IVD INSTRUMENT PRECISION MICROFLUIDIC MANIFOLD BLOCK
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Medical Grade PEEK (Polyetheretherketone) / Austenitic Stainless Steel 316L (UNS S31603)		DWG NO. W3-MED-IVD-MANIFOLD-5.3 / A3
				WEIGHT: ~115.8 g (PEEK) / ~655.0 g (316L)	SCALE: NTS	SHEET: 1 OF 1